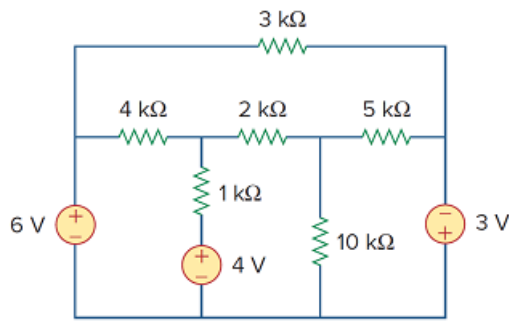


Problem

Determine the current through the $10\text{-k}\Omega$ resistor in the circuit of Fig. 3.93 using mesh analysis.

Figure 3.93:



Step-by-step solution

Step 1 of 12

Consider the following circuit diagram to determine the current through the $10\text{ k}\Omega$ resistor.

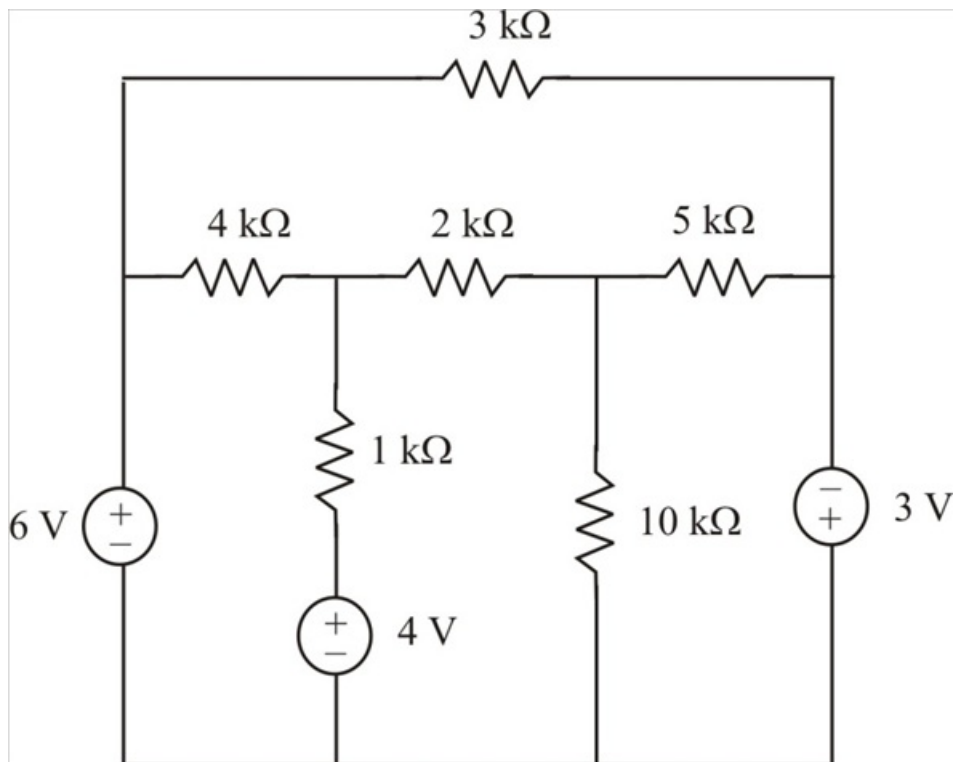


Figure 1

Step 2 of 12

Label the circuit with node voltages and mesh currents as shown in Figure 2.

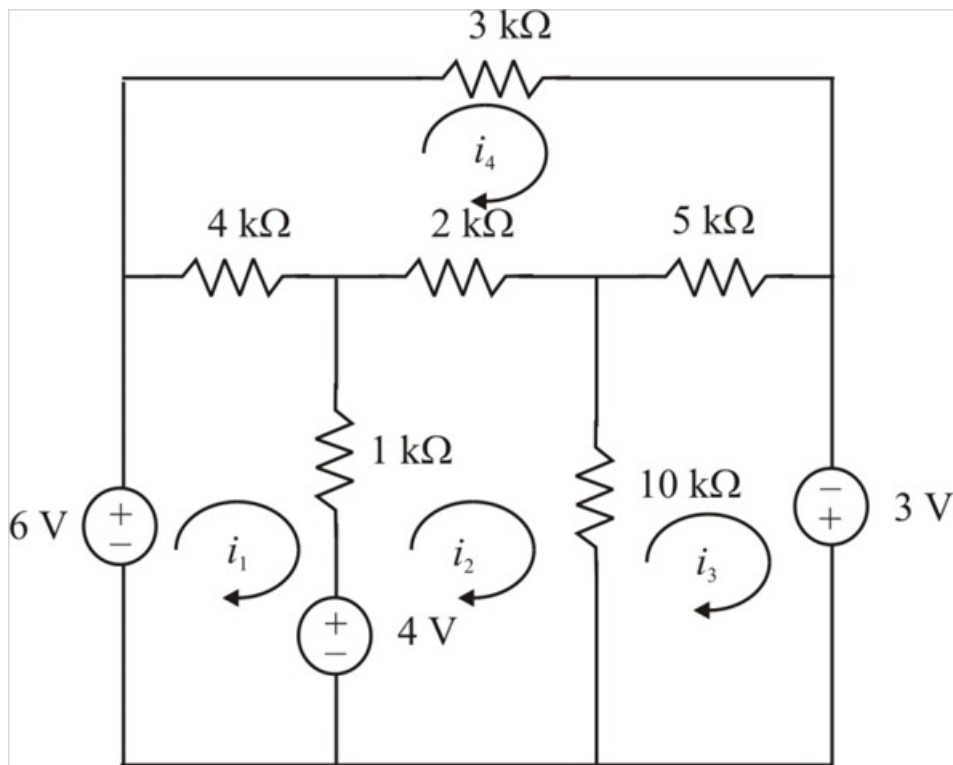


Figure 2

Step 3 of 12

Kirchhoff's voltage law states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Apply Kirchhoff's voltage law to the loop 1 of circuit in Figure 2 to get the following expression:

$$\begin{aligned} -6 \text{ V} + (4 \text{ k}\Omega)(i_1 - i_4) + (1 \text{ k}\Omega)(i_1 - i_2) + 4 \text{ V} &= 0 \\ -6 \text{ V} + (4 \text{ k}\Omega)i_1 - (4 \text{ k}\Omega)i_4 + (1 \text{ k}\Omega)i_1 - (1 \text{ k}\Omega)i_2 + 4 \text{ V} &= 0 \\ (5 \text{ k}\Omega)i_1 - (1 \text{ k}\Omega)i_2 - (4 \text{ k}\Omega)i_4 &= 2 \text{ V} \end{aligned}$$

Assume that the mesh currents are in mA.

$$5i_1 - i_2 - 4i_4 = 2 \text{ V} \dots\dots (1)$$

Step 4 of 12

Apply Kirchhoff's voltage law to the loop 2 of circuit in Figure 2 to get the following expression:

$$\begin{aligned} -4 \text{ V} + (1 \text{ k}\Omega)(i_2 - i_1) + (2 \text{ k}\Omega)(i_2 - i_4) + (10 \text{ k}\Omega)(i_2 - i_3) &= 0 \\ -4 \text{ V} + (1 \text{ k}\Omega)i_2 - (1 \text{ k}\Omega)i_1 + (2 \text{ k}\Omega)i_2 - (2 \text{ k}\Omega)i_4 + (10 \text{ k}\Omega)i_2 - (10 \text{ k}\Omega)i_3 &= 0 \\ -(1 \text{ k}\Omega)i_1 + (13 \text{ k}\Omega)i_2 - (10 \text{ k}\Omega)i_3 - (2 \text{ k}\Omega)i_4 &= 4 \text{ V} \end{aligned}$$

Assume that the mesh currents are in mA.

$$-i_1 + 13i_2 - 10i_3 - 2i_4 = 4 \text{ V} \dots\dots (2)$$

Step 5 of 12

Apply Kirchhoff's voltage law to the loop 3 of circuit in Figure 2 to get the following expression:

$$\begin{aligned}(10 \text{ k}\Omega)(i_3 - i_2) + (5 \text{ k}\Omega)(i_3 - i_4) - 3 \text{ V} &= 0 \\(10 \text{ k}\Omega)i_3 - (10 \text{ k}\Omega)i_2 + (5 \text{ k}\Omega)i_3 - (5 \text{ k}\Omega)i_4 - 3 \text{ V} &= 0 \\-(10 \text{ k}\Omega)i_2 + (15 \text{ k}\Omega)i_3 - (5 \text{ k}\Omega)i_4 &= 3 \text{ V}\end{aligned}$$

Assume that the mesh currents are in mA.

$$-10i_2 + 15i_3 - 5i_4 = 3 \text{ V} \dots\dots (3)$$

Step 6 of 12

Apply Kirchhoff's voltage law to the loop 4 of circuit in Figure 2 to get the following expression:

$$\begin{aligned}(3 \text{ k}\Omega)i_4 + (5 \text{ k}\Omega)(i_4 - i_3) + (2 \text{ k}\Omega)(i_4 - i_2) + (4 \text{ k}\Omega)(i_4 - i_1) &= 0 \\(3 \text{ k}\Omega)i_4 + (5 \text{ k}\Omega)i_4 - (5 \text{ k}\Omega)i_3 + (2 \text{ k}\Omega)i_4 - (2 \text{ k}\Omega)i_2 + (4 \text{ k}\Omega)i_4 - (4 \text{ k}\Omega)i_1 &= 0 \\-(4 \text{ k}\Omega)i_1 - (2 \text{ k}\Omega)i_2 - (5 \text{ k}\Omega)i_3 + (14 \text{ k}\Omega)i_4 &= 0\end{aligned}$$

Assume that the mesh currents are in mA.

$$-4i_1 - 2i_2 - 5i_3 + 14i_4 = 0 \dots\dots (4)$$

Step 7 of 12

Multiply equation (4) by the factor (-2) and add the result to equation (2).

$$\begin{array}{r} -i_1 + 13i_2 - 10i_3 - 2i_4 = 4 \\ (-2)(-4i_1 - 2i_2 - 5i_3 + 14i_4) = 0 \\ \hline -i_1 + 13i_2 - 10i_3 - 2i_4 = 4 \\ 8i_1 + 4i_2 + 10i_3 - 28i_4 = 0 \\ \hline 7i_1 + 17i_2 + 0 - 30i_4 = 4 \end{array}$$

Therefore,

$$7i_1 + 17i_2 - 30i_4 = 4 \text{ V} \dots\dots (5)$$

Multiply equation (5) by the factor (5) and equation (1) by the factor (-7) and add the results.

$$\begin{array}{r} (5)(7i_1 + 17i_2 - 30i_4) = (5)(4 \text{ V}) \\ (-7)(5i_1 - i_2 - 4i_4) = (-7)(2 \text{ V}) \\ \hline 35i_1 + 85i_2 - 150i_4 = 20 \text{ V} \\ -35i_1 + 7i_2 + 28i_4 = -14 \text{ V} \\ \hline 0 + 92i_2 - 122i_4 = 6 \text{ V} \end{array}$$

Therefore,

$$92i_2 - 122i_4 = 6 \text{ V} \dots\dots (6)$$

Step 8 of 12

Multiply equation (2) by the factor (4) and equation (4) by the factor (-1) and add the results.

$$\begin{array}{r} (4)(-i_1 + 13i_2 - 10i_3 - 2i_4) = 4 \text{ V} \\ (-1)(-4i_1 - 2i_2 - 5i_3 + 14i_4) = 0 \\ \hline -4i_1 + 52i_2 - 40i_3 - 8i_4 = 16 \text{ V} \\ 4i_1 + 2i_2 + 5i_3 - 14i_4 = 0 \\ \hline 0 + 54i_2 - 35i_3 - 22i_4 = 16 \text{ V} \end{array}$$

Therefore,

$$54i_2 - 35i_3 - 22i_4 = 16 \text{ V} \dots\dots (7)$$

Step 9 of 12

Multiply equation (7) by the factor (15) and equation (3) by the factor (35) and add the results.

$$\begin{array}{r} (15)(54i_2 - 35i_3 - 22i_4) = (15)(16 \text{ V}) \\ (35)(-10i_2 + 15i_3 - 5i_4) = (35)(3 \text{ V}) \\ \hline 810i_2 - 525i_3 - 330i_4 = 240 \text{ V} \\ -350i_2 + 525i_3 - 175i_4 = 105 \text{ V} \\ \hline 460i_2 + 0 - 505i_4 = 345 \text{ V} \end{array}$$

Therefore,

$$460i_2 - 505i_4 = 345 \text{ V} \dots\dots (8)$$

Step 10 of 12

Multiply equation (6) by the factor (-5) and add the result to equation (8).

$$\begin{array}{r} (-5)(92i_2 - 122i_4) = (-5)(6 \text{ V}) \\ 460i_2 - 505i_4 = 345 \text{ V} \\ \hline -460i_2 + 610i_4 = -30 \text{ V} \\ \hline 460i_2 - 505i_4 = 345 \text{ V} \\ 0 + 105i_4 = 315 \text{ V} \end{array}$$

Simplify the expression to calculate i_4 .

$$\begin{aligned} i_4 &= \frac{315 \text{ V}}{105} \\ &= 3 \text{ mA} \quad (\text{since, the currents are assumed to be in mA}) \end{aligned}$$

Step 11 of 12

Substitute 3 for i_4 in equation (6).

$$\begin{aligned} 92i_2 - 122i_4 &= 6 \text{ V} \\ 92i_2 - 122(3) &= 6 \text{ V} \\ 92i_2 - 366 \text{ V} &= 6 \text{ V} \\ 92i_2 &= 372 \text{ V} \end{aligned}$$

Simplify the expression to calculate i_2 .

$$\begin{aligned} i_2 &= \frac{372 \text{ V}}{92} \\ &= 4.0434 \text{ mA} \quad (\text{since, the currents are assumed to be in mA}) \end{aligned}$$

Substitute 4.04 for i_2 and 3 for i_4 in equation (3).

$$-10i_2 + 15i_3 - 5i_4 = 3 \text{ V}$$

$$-10(4.0435) + 15i_3 - 5(3) = 3 \text{ V}$$

$$-40.435 \text{ V} + 15i_3 - 15 \text{ V} = 3 \text{ V}$$

$$15i_3 - 55.435 \text{ V} = 3 \text{ V}$$

Simplify the expression to calculate i_3 .

$$15i_3 = 58.435 \text{ V}$$

$$i_3 = \frac{58.435 \text{ V}}{15}$$

$$= 3.895 \text{ mA}$$

Calculate the current through the $10 \text{ k}\Omega$ resistance.

$$i_{10 \text{ k}\Omega} = i_2 - i_3$$

$$= 4.0435 \text{ mA} - 3.895 \text{ mA}$$

$$\square 0.148 \text{ mA}$$

$$\square 148 \mu\text{A}$$

Therefore, the current through the $10 \text{ k}\Omega$ resistor is $\boxed{148 \mu\text{A}}$.